PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR NORTH WESTERN ZONE MOCK EXAMINATION SEP. 2024

032/2A CHEMISTRY 2A (ACTUAL PRACTICAL 2A)

2:30HRS **25/9/2024 WEDNESDAY**

INSTRUCTIONS

i. This paper consists of **TWO (2)** questions
ii. Answer question in this paper
iii. Each question carries **twenty-five (25**) marks
iv. Cellular phone and unauthorized material are not allowed in the examination room.
v. Mathematical tables and non- programmable calculators may be used in the examination room.
vi. You may use the following constants
Atomic mass
H=1, C=12, O=16, Na= 23, S=32, K= 39
1dm^3 = 1 Liter = 1000cm^3.

1. You are provided with the following solutions
JJ. A solution containing 3.0g of acetic in 0.5dm^3 of solution
KK. A solution containing 1.5g of impure potassium hydroxide in 250cm^3 of solution.
MM: Phenolphthalein indicator

PROCEDURE

i) Pipette 20cm^3 or 25cm^3 of solution KK into clean and dry conical flask.
ii) Add to the same conical flask about three drops of solution MM and shake the mixture well.
iii) Put solution JJ into the burette and record the initial volume of the burette.
iv) Titrate solution JJ from the burette against solution in the conical flask until colour change is observed.
v) Records the final volume of the burette
vi) Repeat procedure (i- v) to obtain three more readings and tabulate your results in a table as shown below.

Table 1

Volume of pipette used was …….

titration	pilot	1	2	3
Final volume (cm^3)				
Initial volume (cm^3)				

Volume used in (cm^3)				

SUMMARY

……..cm^3 of solution KK required……. Cm^3 of solution JJ under solution mm for complete reaction.

QUESTIONS

(i) Is the use of methyl orange indicator in this titration is as suitable as phenolphthalein indicator?

(ii) Write a well-balanced chemical equation between KK. And JJ

(iii) Calculate the average volume of acid used in this titration.

(iv) Why it is advisable to clean the pipette with distilled water before pipetting solution KK?

(v) Calculate.

a) Concentration of solution KK in $gldm^3$

b) Concentration of solution JJ in $gldm^3$

c) Molarity of solution KK

d) Percentage purity of potassium hydroxide

2. You are provided with solid sample Q that contains one cation and one onion. Carry out the guided qualitative analysis procedures to determine cation and anion present in solid sample Q.

S/N	EXPERMENT	OBSERVATION	INFERENCE
1	Observe the appearance of solid sample Q		
2	Put 0.5g of solid sample Q in a test tube and heat it gently		
3	Put 0.5g of solid sample Q in a test tube and add 3mls of dilute HCL		
4	Put O.5g of solid sample in a clean and dry test tube and add $3cm^3$ of concentrated H_2SO_4. If no reaction warms the content gently		
5	Perfume flame test of solid perform Q		
6	Dissolve a spatula fully of solid sample Q in an appropriate solvent and divide the solution into four portions		
I	To first portion of solution sample Q above, add sodium hydroxide solution drop wise until in excess.		
ii	To the second portion add 2drops of dilute HNO_3 followed by $3cm^3$ of silver nitrate solution, then add excess ammonia solution.		
iii	To the fourth portion of solution of simple Q above, add $3cm^3$ of sodium hydroxide solution just to cover the whole solution and warm the mixture gently. Test for any		

	gas evolved.		

QUESTIONS

(i) The cation present in solid sample Qis ……. And onion present in solid sample a is ……

(ii) Write the formula of solid sample Q

(iii) Write a well-balanced chemical equation that occurs in stage (2) of this experiment.

(iv) Why concentrated hydrochloric acid does is mostly used in flame test during qualitative analysis.